



UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 23560

LM INFERENCE FROM SCIENTIFIC DATA

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. From the definition of the probability of event A and event B , derive the following form of Bayes' equation,

$$P(B|A) = \frac{P(B)}{P(A)} P(A|B).$$

[3]

A more familiar form of Bayes' formula for use in Bayesian inference is,

$$P(\theta|D) = \frac{P(D|\theta) P(\theta)}{P(D)},$$

where θ is the set of model parameters and D is the data.

Identify each of the following terms in the above and give a short explanation of each term in the context of Bayesian inference,

- The Prior probability distribution (prior)
- The Likelihood
- The Posterior probability distribution (posterior)
- The Evidence

[4]

In the following, you will use a likelihood function that is the Poisson distribution probability density function (PDF),

$$P(n|\lambda) = \frac{\exp(-\lambda) \lambda^n}{n!},$$

where n is the number of occurrences, and λ is a parameter that is greater than zero. The prior you will use is the PDF of the Gamma distribution,

$$f(\lambda|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} \exp(-\beta\lambda),$$

where $\Gamma(u)$ is the gamma function of u .

Write down Bayes' equation using the above likelihood and prior and simplify. You may write the evidence as $\mathcal{Z}$.

[3]

Show that,

$$\mathcal{Z} = \frac{\beta^\alpha \Gamma(n + \alpha)}{(\beta + 1)^{n+\alpha} \Gamma(\alpha) n!}.$$

For full marks ensure you show your working.

[15]

In an experiment, data D is collected and the evidences for 2 models, M_1 and M_2 , with parameters θ_1 and θ_2 respectively are found. Write down the posterior odds ratio for the two models and define all terms used.

It is found that posterior odds ratio is exactly unity - write down an expression for your posterior belief and comment on how the experiment should report the posterior probability distribution for a particular parameter that is found in both θ_1 and θ_2 .

[5]

2. The student's t -distribution has the following probability density function (PDF):

$$f_{\nu}(t) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\nu\pi} \Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

where $\Gamma()$ is the gamma function, ν is the degrees of freedom, and t is in the independent variable.

Write down the definitions for the mean, $\langle x \rangle$, and the variance, $\text{Var}(x)$ for a continuous, real-valued random variable x with PDF $P(x)$. Comment on the limits of the integrals in your definitions.

[3]

Sketch the student's t -distribution PDF with $\nu = 3$ and the PDF of the Gaussian distribution with mean of zero and variance of unity on the same plot. Label your axes. If you are concerned that your sketch is inaccurate you can, in addition, state in words the important differences between the two curves.

[5]

Find the mean of the student's t -distribution, $\langle t \rangle$. Explain your reasoning.

[4]

Show that the variance of the Student's t -distribution with $\nu > 2$ degrees of freedom is given by $\text{Var}(t) = \frac{\nu}{\nu-2}$.

Note that

$$\frac{\Gamma(\alpha + 1)}{\Gamma(\alpha)} = \alpha,$$

and the standard integral required is,

$$\int_0^{\infty} \frac{x^2}{(1 + x^2)^{\alpha}} dx = \frac{\sqrt{\pi}}{4} \frac{\Gamma(\alpha - \frac{3}{2})}{\Gamma(\alpha)}.$$

[18]

3. Suppose there is a beta prior distribution, $\text{Beta}(\alpha, \beta)$ on the probability θ that a coin will land heads when spun in a specified manner.

Write down and simplify the posterior probability distribution, up to a proportionality constant $\mathcal{Z}$, of getting k heads if the coin is spun n times. **[5]**

The probability density function (PDF) for the beta distribution is,

$$\text{Beta}(\theta|\alpha, \beta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)},$$

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)},$$

and $\Gamma(u)$ is the Gamma function, which for positive integers is $\Gamma(n) = (n-1)!$.

The probability mass function (PMF) for the binomial distribution is,

$$P(\theta|n, k) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}.$$

Using a prior of $\text{Beta}(4, 4)$ show that the exact posterior distribution is,

$$P(\theta|n, k) = \frac{\Gamma(n+8)}{\Gamma(4+k) \Gamma(4+n-k)} \theta^{3+k} (1 - \theta)^{3+n-k}.$$

[5]

Still with a prior that is $\text{Beta}(4, 4)$ on θ , derive the posterior probability distribution when $n = 10$ spins and $k \leq 2$ heads. That is you know that in 10 spins there were less than three heads but you do not know how many less than three. You may use an unnormalised posterior distribution with an unknown constant of proportionality. **[8]**

For a further 2 marks, find the value of your constant of proportionality and write down the exact posterior. **[2]**

Sketch the posterior distribution for θ , with labels on both axes and indicative numbers on the θ axis. Calculate the mean of the posterior for $n = 10$ spins and $k \leq 2$ heads and show this on the sketch relative to where you draw the mode of the distribution.

If you need to calculate the mean of a beta distribution, it might be useful to know that $\Gamma(x+1) = \Gamma(x)x$. **[10]**

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.